Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

Question permutation table:

Version		1	2	3	4
Question					
1	→	4	5	1	8
2	→	3	9	2	9
3	→	9	2	4	2
4	→	7	6	7	5
5*	→	8	7	8	6
6	→	1	8	5	3
7	→	2	4	6	1
8	→	6	3	9	7
9	→	5	1	3	4

*: Question follows previous

1. The following 19 test scores are based on a random sample of final exam scores (sorted in increasing order) for the population of students who took an elementary statistics course at a large university over a 5 year period:
   61 65 69 70 71 73 75 77 77 82 82 83 84 85 86 89 92 92 94
   Suppose the largest test score in the sample changed from a 94 to a 90. Describe how each of the following statistics would change.
   The median would ___(1)___.
   The mean would ___(2)___.
   The standard deviation would ___(3)___.

   (a) decrease, decrease, decrease
   (b) remain the same, decrease, increase
   (c) remain the same, decrease, decrease
   (d) decrease, increase, remain the same

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

2. Suppose a population of test scores is skewed right. What would be the shape of the sampling distribution for sample mean test scores from samples of 50 students?

   (a) Skewed right
   (b) Skewed left
   (c) Approximately normal
   (d) Unable to determine

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)

3. If a bootstrap distribution of 1,000 sample medians was generated rather than 1,000 sample means, what would be the shape of the bootstrap distribution of sample medians?

   (a) Approximately normal
   (b) Skewed right
   (c) Skewed left
   (d) Unable to determine

   Correct answers: (d)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(b)	(b)	(c)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(d)	(c)	(d)	(c)

Noxious weeds are invasive plants that quickly spread and deplete the soil of nutrients that are required for all aspects of farming and ranching. With much at stake, the state of Montana has hired a team of researchers to explore the issue. Researchers want to know the average number of noxious weeds per acre in North Central Montana. To do so, the researchers divided the area of interest into 1,000,000 acres and randomly selected 24 of these acres. For each selected acre, the noxious weeds were counted and removed. The sample mean was 103.47 weeds, and the sample standard deviation was 46.034 weeds.

4. What is the population of interest?

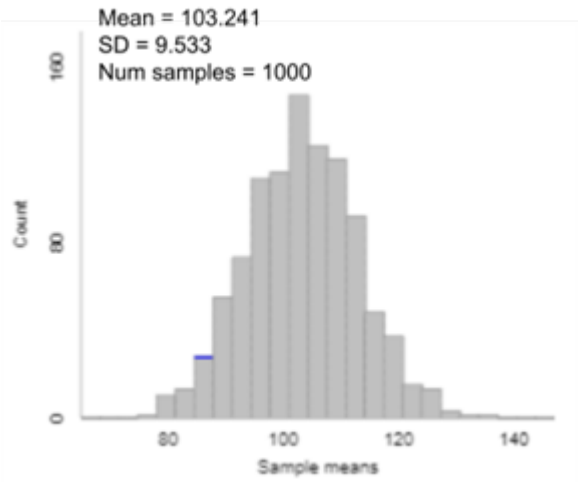
   (a) All acres in North Central Montana
   (b) The 24 selected acres in the sample
   (c) The true mean number of noxious weeds per acre in North Central Montana
   (d) The sample mean number of noxious weeds per acre

   Correct answers: (a)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(c)	(b)	(b)	(c)
(d)	→	(b)	(d)	(a)	(b)

5. Below is the simulated bootstrap distribution of 1,000 sample means created from resampling the observed data.



How was one bootstrapped sample generated in the plot above?

(a) Randomly select 24 acres from the area of interest and calculate the mean number of noxious weeds.
(b) Randomly select 24 acres from the acres selected for the sample, without replacement, and calculate the mean number of noxious weeds.
(c) Randomly select 24 acres from the acres selected for the sample, with replacement, and calculate the mean number of noxious weeds.
(d) Randomly select 24 acres from a hypothetical population of acres and calculate the mean number of noxious weeds.

Correct answers: (c)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(c)	(b)	(d)	(d)
(c)	→	(d)	(d)	(a)	(b)
(d)	→	(b)	(c)	(b)	(c)

6. Suppose a confidence interval is constructed using repeated tests of significance to develop an interval of plausible values. Two-sided tests are used each time with the following null hypotheses and resulting p-values.

Null	p-value	Null	p-value
$\pi = 0.45$	0.009	$\pi = 0.53$	0.787
$\pi = 0.46$	0.032	$\pi = 0.54$	0.572
$\pi = 0.47$	0.062	$\pi = 0.55$	0.373
$\pi = 0.48$	0.126	$\pi = 0.56$	0.142
$\pi = 0.49$	0.371	$\pi = 0.57$	0.077
$\pi = 0.50$	0.598	$\pi = 0.58$	0.042
$\pi = 0.51$	0.733	$\pi = 0.59$	0.021
$\pi = 0.52$	0.986	$\pi = 0.60$	0.003

Using the results, give a 95% confidence interval.

(a) (0.50, 0.54)
(b) (0.47, 0.57)
(c) (0.45, 0.60)
(d) (0.46, 0.58)

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(d)	(c)
(b)	→	(d)	(c)	(a)	(b)
(c)	→	(b)	(d)	(c)	(d)
(d)	→	(a)	(b)	(b)	(a)

7. Commercial fisherman working certain parts of the Atlantic Ocean sometimes find efforts being hindered by the presence of whales. Ideally, the whales could be scared away without frightening the fish. Experience has shown that whales leave the sight of a fishing boat 40% of the time. One of the strategies being experimented with is to transmit underwater sounds of a killer whale. On the 52 occasions that this technique has been tried, the technique worked 24 times (that is, the whales immediately left the area). A 99% confidence interval for π is $(0.29, 0.65)$. How would you interpret this interval?

(a) When underwater sounds of killer whales are transmitted, fishermen can be are 99% confident that whales leave the sight of a fishing boat 46% of the time.
(b) When underwater sounds of killer whales are transmitted, fishermen can be 99% confident that whales leave the sight of a fishing boat between 29% and 65% of the time.
(c) 99% of whales will leave the sight of a fishing boat if the whales hear sounds of a killer whale.
(d) If another sample of 52 occasions is collected, a 99% chance exists that the sample proportion of whales who leave the sight of a fishing boat will be between 0.29 and 0.65.

Correct answers: (b)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(d)	(a)	(b)
(b)	→	(c)	(a)	(b)	(a)
(c)	→	(d)	(c)	(d)	(d)
(d)	→	(b)	(b)	(c)	(c)

8. Let π be some population proportion of interest and suppose a 99% confidence interval for π is calculated to be (0.32, 0.41) and a 95% confidence interval is calculated to be (0.34, 0.39). Also, suppose the hypothesis test is $H_0 : \pi = 0.33$ and $H_A : \pi \neq 0.33$. What can be said about the corresponding p-value for this test?

   (a) The corresponding p-value will be less than 0.01.
   (b) The corresponding p-value will be between 0.01 and 0.05.
   (c) The corresponding p-value will be between 0.05 and 0.10.
   (d) The corresponding p-value will be greater than 0.10.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(a)	(d)
(b)	→	(d)	(c)	(d)	(b)
(c)	→	(c)	(a)	(c)	(c)
(d)	→	(a)	(d)	(b)	(a)

9. Suppose a test of significance is conducted with $H_0 : \pi = 0.50$ and $H_A : \pi \neq 0.50$ with a resulting p-value of 0.07. Also, 99% and 95% confidence intervals are constructed from the data. What will be true about the confidence intervals?

   (a) Both the 99% and the 95% intervals will contain 0.50.
   (b) Neither the 99% nor the 95% intervals will contain 0.50.
   (c) The 99% interval will contain 0.50, but the 95% interval will not contain 0.50.
   (d) The 99% interval will not contain 0.50, but the 95% interval will contain 0.50.

   Correct answers: (a)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(a)	(c)	(d)
(b)	→	(b)	(c)	(d)	(b)
(c)	→	(d)	(b)	(b)	(a)
(d)	→	(a)	(d)	(a)	(c)